

# Gravitation

MP

Teach You Physics Hand In Hand

March 2026

## 1 SOLUTION to Problem in EM post

- Thanks to Oscar Chan for proposing this question.
- Level of difficulty: **5/10**

a) The equation of motion reads:

$$mg + F_B - \beta v - F_{\text{induced}} = ma$$

where

$$F_B = \frac{BV_0\ell}{R}, \quad F_{\text{induced}} = BI_{\text{induced}}\ell = \frac{B^2\ell^2v}{R}$$
$$\Rightarrow a = g + \frac{BV_0\ell}{mR} - \frac{\beta v}{m} - \frac{B^2\ell^2v}{mR}$$

In terms of  $\ddot{y} = \frac{dv}{dt}$  and  $\dot{y} = v$ :

$$\ddot{y} = -\left(\frac{\beta}{m} + \frac{B^2\ell^2}{mR}\right)\dot{y} + g + \frac{BV_0\ell}{mR} \quad [4 \text{ points}]$$

Let

$$\ddot{y} + \alpha\dot{y} = \gamma, \quad \alpha = \frac{\beta}{m} + \frac{B^2\ell^2}{mR}, \quad \gamma = g + \frac{BV_0\ell}{mR}$$

By the integrating factor method:

$$\frac{d}{dt}(e^{\alpha t}\dot{y}) = \gamma e^{\alpha t}$$

Applying  $\dot{y}(0) = 0$ :

$$e^{\alpha t}\dot{y} = \frac{\gamma}{\alpha}(e^{\alpha t} - 1)$$
$$\dot{y} = \frac{\gamma}{\alpha}(1 - e^{-\alpha t}) \quad [4 \text{ points}]$$

Further integrating:

$$y(t) - y_0 = \frac{\gamma}{\alpha} \int (1 - e^{-\alpha t}) dt$$

$$y(t) = y_0 + \frac{\gamma}{\alpha}t + \frac{\gamma}{\alpha^2}(e^{-\alpha t} - 1) \quad [2 \text{ points}]$$

b) We have  $v = pv_T$ , where  $v_T = v(t \rightarrow \infty) = \frac{\gamma}{\alpha}$  [2 points] and  $p = \frac{x}{100}$ . Then

$$e^{-\alpha t_1} = 1 - p$$

$$t_1 = -\frac{\ln(1-p)}{\alpha} \quad [4 \text{ points}]$$

From part (a), we have:

$$y(t_1) = y_0 - \frac{\gamma}{\alpha^2} \ln(1-p) + \frac{\gamma}{\alpha^2}((1-p) - 1)$$

$$\Delta y = -\frac{\gamma}{\alpha^2}(\ln(1-p) + p)$$

$$\Delta y = -\frac{\gamma}{\alpha^2} \left( \ln \left( 1 - \frac{x}{100} \right) + \frac{x}{100} \right) \quad [4 \text{ points}]$$

This is a very common classical model that appears in DSE frequently.

## 2 Trajectories (pathes of motion) of Planets

The orbit of the Earth around the Sun is in fact not a circle. It is an **ellipse**. Gravitational force is given by:

$$\vec{F}(r) = -\hat{\mathbf{r}} \frac{GMm}{r^2}$$

$$\Rightarrow U(r) = \int_r^\infty \frac{GMm}{r^2} dr = -\frac{GMm}{r}$$

By conservation of **angular momentum**  $L$  and mechanical energy:

$$\frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 - \frac{GMm}{r} = E$$

$$\frac{1}{2}m \left( \frac{dr}{d\theta} \frac{L}{mr^2} \right)^2 + \frac{1}{2}mr^2 \left( \frac{L}{mr^2} \right)^2 - \frac{GMm}{r} = E$$

After some algebra:

$$\begin{aligned} \Rightarrow r(\theta) &= \frac{L^2}{GMm^2 \left[ 1 + \sqrt{1 + \frac{2EL^2}{G^2M^2m^3}} \cos(\theta - \theta_0) \right]} \\ \Rightarrow r(\theta) &= \frac{r_0}{1 + e \cos \theta} \end{aligned}$$

where:

- $r_0 = \frac{L^2}{GMm^2}$
- **Eccentricity**  $e = \sqrt{1 + \frac{2EL^2}{G^2M^2m^3}}$
- $E$  is total energy of the Earth orbit, which is a constant.

In fact, the eccentricity of Earth orbit can be approximated to be 0. We can show that under this approximation, the ellipse orbit is nearly a circle with  $r_0 \approx$  radius of orbit.

### 3 Problem

Mathematically, for any **scalar field**  $\Psi(r, \phi, z)$ , we can convert it into a **vector field**  $\vec{\Psi}(r, \phi, z)$  by an operation called **gradient**  $\nabla$ :

$$\nabla\Psi(r, \phi, z) = \frac{\partial\Psi}{\partial r}\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial\Psi}{\partial\phi}\hat{\phi} + \frac{\partial\Psi}{\partial z}\hat{\mathbf{z}} = \vec{\Psi}(r, \phi, z)$$

#### Part A. Introduction

ai) [1 point] As a simple warm-up, prove that the negative gradient of gravitational potential equals to gravitational field. i.e.  $\vec{g}(r) = -\nabla V(r)$

Given that there is potential (scalar) field  $V(r)$  in a space, where  $r^2 = x^2 + y^2$ . The corresponding central force does not obey the inverse-square law. Instead, an inverse -  $p$  - power law with proportional constant  $k$  is obeyed, where  $p > 0$ . Denote the effective "mass" of an object under this field be  $m$ .

aii) [1 point] Derive the expression of  $V(r)$ .

aiii) [3 points] Find the time required for an object with effective mass  $m$  to hit the "source" of the field in this space with initial separation  $y_0\hat{y}$ . Assume the source is located at the origin. Leave your answer as an integral.

#### Part B. Classical application

Suppose now that we introduce a system called the "Gravitational Slingshot" into this field, which means using the field to set an object into desired motions. The separation between  $m$  and the "source" is now  $y_0\hat{y} + x_0\hat{x}$ , and the initial velocity of  $m$  is  $v_0\hat{y}$ , where  $x_0 > 0$ . The total energy of the system is  $E$ . Assume the velocity is too large that the motion is unbounded and the mass will turn around the "source".

bi) [2 points] Find the final speed of the mass. State the condition for the initial velocity such that the motion is unbounded.

bii) [6 points] In polar coordinate system, solve for the trajectory  $\theta(\frac{1}{r})$ . Leave your answer as an integral.

### Part C. Quantum view

In this part, you may find Schrödinger's Equation helpful:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

The space is filled with many infinitesimal masses of mass  $\delta$ . Suppose that the potential field changes over time. It obeys an evolution formula:

$$\frac{\partial V}{\partial t} + \alpha V = 0$$

where  $\alpha$  is a constant.

ci) [1 point] Given the Poisson's equation describes the relation between the field and the density of the masses:

$$\nabla^2 V(r, t) \propto \rho(r, t)$$

Derive the expression for the density of infinitesimal masses  $\rho(r, t)$ . Use  $\rho(r, 0)$ ,  $\alpha$  and  $t$  to express your answer. (\*It's an easy question!!!)

cii) [6 points] At  $t=0$ , the potential  $V(r, 0)$  is same as that derived in Part A. Assume a mass  $\delta$  is performing some orbital motion with angular momentum  $L$ . Given that the effective potential reach minimum at  $r = r_0$  and it can be expressed as:

$$V_{\text{eff}}(r, t) = V(r, t) + \frac{\hbar^2 L^2}{2\delta r^2}$$

The wave function of the infinitesimal mass  $\delta$  is:

$$\Psi(r, t) = \left(\frac{\sigma}{\pi}\right)^{1/4} A(t) e^{-\sigma B(t)(r-r_0)^2/2}$$

where  $\sigma$  is some constant.

Derive  $A(t)$  and  $B(t)$ . Leave your answer as differential equation(s).

The solution will be uploaded in the next post and MP love you all :)